

Definitions

Hilbert order: qubit tensor cavity

Active basis: ($|g,0\rangle$, $|e,0\rangle$, $|g,1\rangle$, $|e,1\rangle$, ...)

N_{active} : number of addressed Fock manifolds in the target operator

Strict criterion: exact blockwise $SU(2)$ agreement

Relaxed criterion: per-block left Z-gauge equivalence

State tests: $|g\rangle$, $|e\rangle$, $+/-x$, $+/-y$ per active block + cross-block cavity superposition